

RADIATION DAMAGE EFFECTS IN OXIDE DISPERSION STRENGTHENED STEEL ALLOYS

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ABSTRACT

Oxide dispersed strengthened steel (ODSS) is widely used in nuclear industry due to its excellent creep and void swelling resistance. The presence of high density of uniformly distributed nano clusters in steel alloys can reduce both void swelling and radiation embrittlement in current reactors. ODSS alloys also show radiation stability at high temperatures anticipated in advanced reactors. Since fabrication and characterization is still immature and need extended research by integrated multi-scale simulation and experimental studies. Computational approaches like Molecular Dynamics (MD) can help to understand mechanism for radiation damage in ODSS. Atomistic to bulk properties can be analyzed through multi-scale simulations. It is recommended to combine multi-scale modeling approach with experimental studies validation. This article provides a comprehensive perspective of radiation damage effects covering both experimental and simulation methods. More discussion is based on simulation performed through MD for understanding radiation damage in ODS alloys. MD simulations for irradiation stability, damage cascades and strengthening mechanism are also discussed.

1. INTRODUCTION

Modern world's drastic energy requirements, demands the exploration of various alternate energy resources. Nuclear energy is being regarded as an alternative, to serve this purpose (Allen *et al.*, 2010). Nuclear fuels for current designs of advanced fission and fusion reactors require radiation resistant structural and fuel materials to withstand harsh environments. Since, irradiation at higher stress, corrosive

environments and temperature causes aging and material failure. Severe radiation and temperature can definitely change mechanical properties of any material. Microstructural and micro-chemical properties also change due to prolonged radiation exposure. Noticeable physical properties change include degradation phenomenon such as: Radiation Assisted Growth (RAG), Radiation Induced Swelling (RIS), Corrosion Stress Cracking (CRS), embrittlement, hardness. Thus, mainly affecting the structural material (Kenik and Busby, 2012). Generation IV reactors are the advanced reactor's technology claimed to have improved energy deployment, safety and reliability (Duan *et al.*, 2017). They are considered as being more economical and having a longer lifetime. Their high safety criteria require material with exceptional mechanical properties and durability, which ODSS can serve this purpose because of their irradiation stability and creep resistance (Ukai and Fujiwara, 2002).

Since stainless steels (SS) with 10.5% Cr is an excellent corrosive resistant material. It is being used in nuclear containment, cladding materials and floor systems for decades. SS is categorized as Austenitic steel, Ferritic, Martensitic duplex steels (dual phase alloy being mixture of ferrite and austenite). Austenite stainless steel (ASS) 304L, 316L is considered being a good choice as structural material for current PWR and BWR. For next generation reactors extended service condition requires high temperature and radiation resistant materials that can be obtained by introduction of nanoscale dispersoids in the steel matrix. The mechanical properties of steel could be improved by